Northeastern Mindanao Colleges
COLLEGE OF INFORMATION TECHNOLOGY
Surigao City, Philippines

INTEGRATED DIGITAL SYSTEM FOR BARANGAY POBLACION HEALTH CENTER SERVICES AND COMMUNITY NOTIFICATIONS

A Project Proposal
Presented to the
Faculty of College of Information Technology
Northeastern Mindanao Colleges
Surigao City

In partial fulfilment
Of the Requirements for System Analysis and Design

By:

Baluya, Jerald P.
Embodo, Sherwyne D.
Mapalad, Dencil S.
Montenegro, John Mark B.

May 2026

CHAPTER 1

INTRODUCTION

This chapter presents the project context, purpose and description, objectives and the scope and limitation of the study.

Project Context

Barangay health centers serve as the frontline of public healthcare in the Philippines, yet their operational capacity remains largely constrained by outdated, manual processes. Barangay Poblacion Health Center is no exception patient records are maintained on paper logbooks, medicine inventory is tracked manually, and appointments are managed without a structured system. These conditions directly contribute to data retrieval delays, elevated risk of record loss or damage, and recurring medicine stock-outs that disrupt health service delivery at the community level.

The scale of this problem is well-documented at the national level. According to the Philippine Statistics Authority (2020), a significant number of barangay health centers across the country still operate without any form of digital health record system, resulting in systemic inefficiencies in data management and service delivery. The Department of Health (2019) further identified the absence of a centralized, accessible platform for community health notifications as a

persistent and unresolved gap in local health governance. Despite the development of health information systems at the national level, these platforms are designed for large-scale facilities and are not adapted to the localized operational needs of individual barangays.

The problem extends beyond internal operations. Community health communication in channels that are Barangay Poblacion relies on physical posting and word-of-mouth inherently limited in reach and reliability. As a result, critical health announcements such as vaccination schedules, medical missions, and disease advisories frequently fail to reach residents in time, leading to missed health services and reduced community participation in public health programs. This communication gap is particularly severe for residents who lack access to the internet or social media, leaving them with no alternative means of receiving health updates.

To directly address these operational and communication deficiencies, this study proposes the development of the Integrated Digital System for Barangay Poblacion Health Center Services and Community Notifications (IDS-BPHCSN) a web-based platform purpose-built for Barangay Poblacion Health Center. The system consolidates patient records management, medicine inventory tracking, appointment scheduling, and community health notifications into a single, accessible digital platform, eliminating the inefficiencies of manual processes

and bridging the communication gap between the health center and the community.

Purpose and Description

The purpose of this project is to develop a web-based application titled Integrated Digital System for Barangay Poblacion Health Center Services and Community Notifications (IDS-BPHCSN), which helps health center staff and barangay officials manage daily operations more efficiently. A web-based system was chosen over desktop or mobile alternatives because it allows multi-user access across different devices without requiring software installation, making it more practical and cost-effective for barangay-level deployment where hardware resources may be limited. The IDS-BPHCSN replaces manual logbooks with a secure digital database, allowing for quick encoding and retrieval of patient records, consultation history, and medicine inventory.

As the staff utilizes the system, they can easily monitor stock levels and generate reports, while barangay officials can use the platform to broadcast health-related announcements directly to residents. The IDS-BPICSN is both administrative and informational, making it a comprehensive tool for enhancing the health

center's workflow and fostering a well-informed community in Barangay Poblacion.

Objective of the Study

General Objective

The main objective of this project is to develop the Integrated Digital System for Barangay Poblacion Health Center (IDS-BPHCSN), a web-based application that digitizes health center services and inventory management while providing a reliable channel for community notifications.

Specific Objectives

1. To design and develop a patient records management module that eliminates data retrieval delays and minimizes the risk of patient record loss.
2. To develop a medicine and supplies inventory module that prevents stock-outs through automated low-stock alerts and real-time stock level monitoring.
3. To design and develop a scheduling module and a community notification module that improve appointment coordination and expand health information reach via web and SMS.

4. To develop an automated report generation module that enables data-driven decision-making through consolidated health service summaries and inventory status reports.

Scope and Limitation

Scope

- The system manages patient records, including personal information and consultation history.
- The system can monitor and update the inventory of medicines and medical supplies.
- The system sends automated low-stock alerts to health center staff when medicine or supply quantities fall below defined threshold levels.
- The system enables the scheduling of appointments and health center activities.
- The system can post announcements.
- The system includes an SMS notification feature allowing officials to send alerts to basic cellphones to ensure residents without internet access stay informed.
- The system generates reports in terms of medicines distributed, current stock levels, patient consultation summaries, and appointment records

- The application is developed as a web-based system accessible via standard web browsers on both desktop and mobile devices.
- The system is developed using Python with the Flask web framework for backend development, SQLite for database management, Bootstrap 5, HTML, CSS, and JavaScript for the frontend interface, and Semaphore API for SMS notification integration.

Limitation

- The system requires a stable internet connection to access all features and perform operations
- The system does not support offline functionality; all features are inaccessible in the absence of an internet connection.
- The system does not support telemedicine, remote medical diagnosis, or clinical decision-support functionalities.
- The system is limited to Barangay Poblacion Health Center and does not integrate with national health databases such as the Philippine Health Information Exchange.
- The system does not implement enterprise-grade security measures and is governed by the Data Privacy Act of 2012 (Republic Act No. 10173).
- The SMS notification feature is limited to text-based messages only and does not support multimedia content such as images or attachments.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter reviews related literature and studies relevant to the development of the Integrated Digital System for Barangay Poblacion Health Center Services and Community Notifications (IDS-BPHCSN). The review covers both foreign and local studies that support the need for digitizing patient records, inventory management, scheduling, and community. Health notifications at the barangay level. This chapter also presents the synthesis of findings that guided the development of the proposed system.

Foreign

Nathan, Agarwal, Peres, et al. (2024), in “Digital Health Interventions in Primary Care in Low- and Middle-Income Countries,” published in the Journal of Global Health Reports, reviewed eHealth interventions in community-level primary care. The review revealed that national platforms are rarely adapted to individual community health facilities, leaving them without practical digital tools. This gap directly validates the development of IDS-BPHCSN as a localized digital health platform for Barangay Poblacion.

A study by Bhattarai, Shrestha. Xiong, et al. (2022), titled "Strengthening Urban Primary Healthcare Service Delivery Using Electronic Health Technologies," published in Digital Health, examined electronic health tools in urban primary care centers in Nepal. The study found that transitioning to digital records improved data accuracy, reduced retrieval delays, and increased health worker efficiency. These findings support the electronic records management approach adopted in IDS-BPHCSN.

Shaw et al. (2020), in a study indexed in PubMed (PMID: 32672878), demonstrated that a centralized electronic health record system in a rural outreach service significantly reduced clinical errors and improved patient data retrieval speed. The study concluded that integrated digital health records are essential for the safety and efficiency of healthcare in underserved communities. This reinforces the value of the patient records module developed for IDS-BPHCSN.

Tanbeer, and Sykes (2021), in "MyHealthPortal: A Web-Based e-Healthcare Web Portal for Out-of-Hospital Patient Care," published in Digital Health, developed a web portal supporting appointment scheduling, health record monitoring, and

health information sharing. Their system achieved a System Usability Scale score of 92.5%, confirming that weh-based health platforms are effective and

accessible. This supports the web-based design of IDS-BPHCSN for multi-user health center management.

A study on the Web-Based Integrated Public Healthcare Information System of South Korea, published in the National Library of Medicine (PMC3920045), developed a community-level platform for public health centers facing poor data access, weak reporting, and insufficient health program coordination. The centralized web-based system resolved these issues and improved regional health service management. These outcomes align with the goals of IDS-BPHCSN in centralizing health operations for Barangay Poblacion.

Local

Docot. Mirontos, Casuyon, et al. (2025), in “LGU: Health and Sanitation Management System,” published in the Bestlink College of the Philippines Journal of Multidisciplinary Research, developed a digital platform for barangay health records and business permits. The system reduced processing time and improved decision-making through analytics dashboards for barangay officials.

The study is directly comparable to IDS-BPHCSN in replacing manual processes with an integrated digital platform for barangay health operations.

Cortez, Fabian, Bajan, et al. (2025), in “Interactive Health Monitoring System and Accounting Resources Management for Barangay Minuyan Proper Health Center,” published in the Bestlink College of the Philippines Journal of Multidisciplinary Research, developed a web application for patient records, inventory tracking, and health program scheduling. The system reduced reliance on manual documentation and provided real-time access to organized health data, serving as a locally validated precedent for IDS-BPHCSN.

Joshi, Sikder, Mekurja, et al. (2025), in “Digital Health for Barangay Health Workers: A Baseline Assessment in Quezon, Philippines,” published on Research Square, assessed the digital readiness of Barangay Health Workers in Quezon Province. The study found that 94.4% of BHWs owned mobile phones but most identified patient registries and health notifications as critical unmet needs. These findings highlight the same operational and communication gaps addressed by IDS-BPHCSN in Barangay Poblacion.

Elepano, Tan-Lim, Jayelosa, et al. (2025), in “Implementing Electronic Health Records in Philippine Primary Care Settings.” Published in JMIR Medical Informatics, investigated HER adoption across primary care facilities in the Philippines. The study confirmed that Philippine health facilities are operationally ready for digital transformation when systems are Designed to match their workflows. This directly validates the development of IDS-BPICSN as a digital records platform for a Philippine barangay health center.

Pescadero, Saltore, and Toledo (2020), in “Barangay Health Management System,” developed a computerized platform for Barangay Villamonte Health Center, which previously relied on manual processes for patient records, medicine inventory, and scheduling. The system provided a digital database for consultation tracking, inventory management, and report generation for the City Health Office. The study confirmed that barangay health centers greatly benefit from purpose-built digital systems, validating the objectives of IDS-BPHCSN.

Synthesis of the Review

The review of foreign literature establishes that digital health systems improve efficiency, accuracy, and accessibility of primary healthcare at the community level. Nathan et al. (2024) and Bhattarai, et al. (2022) confirmed that localized

tools are essential where national platforms fail individual facilities, while Shaw et al. (2020), Tanbeer and Sykes (2021), and the Korean PHIS study showed that integrated web-based platforms enhance data management, usability, and health service coordination.

Local studies reinforce the same conclusions within the Philippine context, where barangay health centers face challenges from manual processes and poor community communication. Docoj et al. (2025) and Cortez et al. (2025) proved

the feasibility of web-based systems at the barangay level, Joshi et al. (2025) and Elepano et al. (2025) confirmed the urgent need for digital tools in Philippine primary care, and Pescadero et al. (2020) validated that digital systems greatly improve barangay health operations.

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CHAPTER 3

TECHNICAL BACKGROUND

This chapter outlines the technical description of the project, including the conceptual framework, software development methodology, system requirements, system architecture process model and data design, and the project schedule of the Integrated Digital System for Barangay Poblacion Health Center Services and Community Notifications (IDS-BPHCSN)

Conceptual Framework

The IDS-BPHCSN is developed to optimize the management of patient records, medicine inventory, appointment scheduling, and community health notifications at Barangay Poblacion Health Center. Through the Input-Process-Output (IPO) framework, the system demonstrates how essential data such as patient information, medicine stock levels. Appointment details, and health announcements are processed by core modules including records management, inventory tracking, scheduling, and notification dispatch. These processes collectively produce the Integrated Digital System for Barangay Poblacion Health

Center Services and Community Notifications (IDS-BPHCSN) as the primary output, a web based platform designed to promote accuracy, efficiency, and transparency in barangay health center operations.

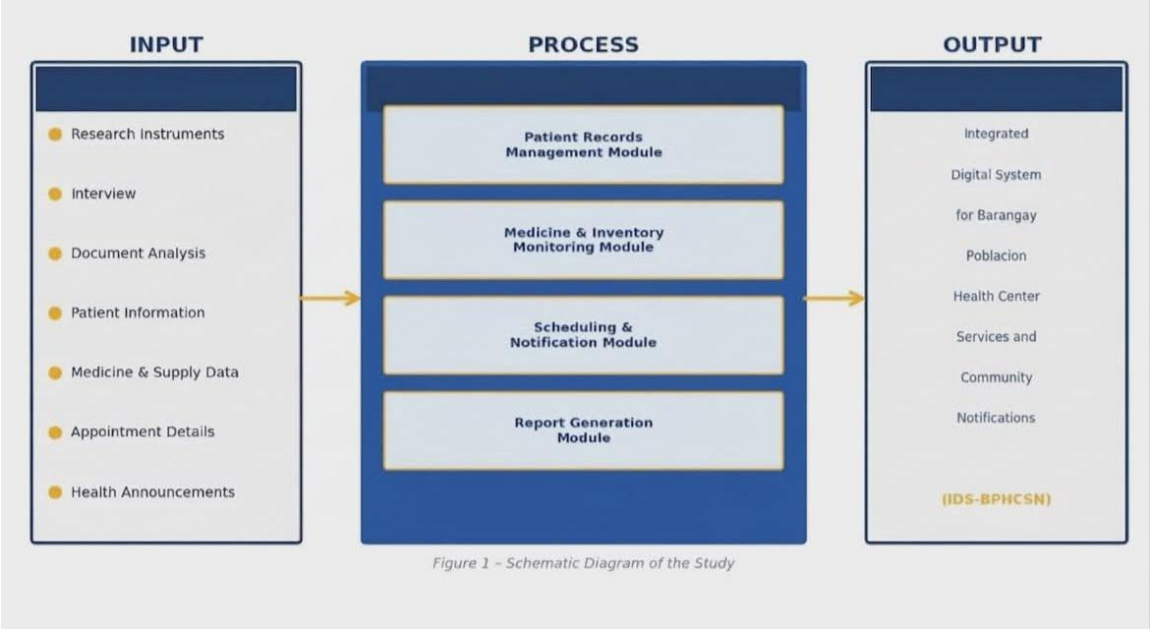


Figure 1: **Schematic Diagram of the Study (IPO Model)**

Methodology

The development of the IDS-BPHCSN followed the Agile Software Development Life Cycle (SDLC), a flexible and iterative methodology that allows for continuous

improvement throughout the development process. Agile was selected because it enables the development team to build the system incrementally one module at a time making it well-suited for a capstone-level project where requirements may evolve through ongoing consultation with the health center staff and barangay officials.

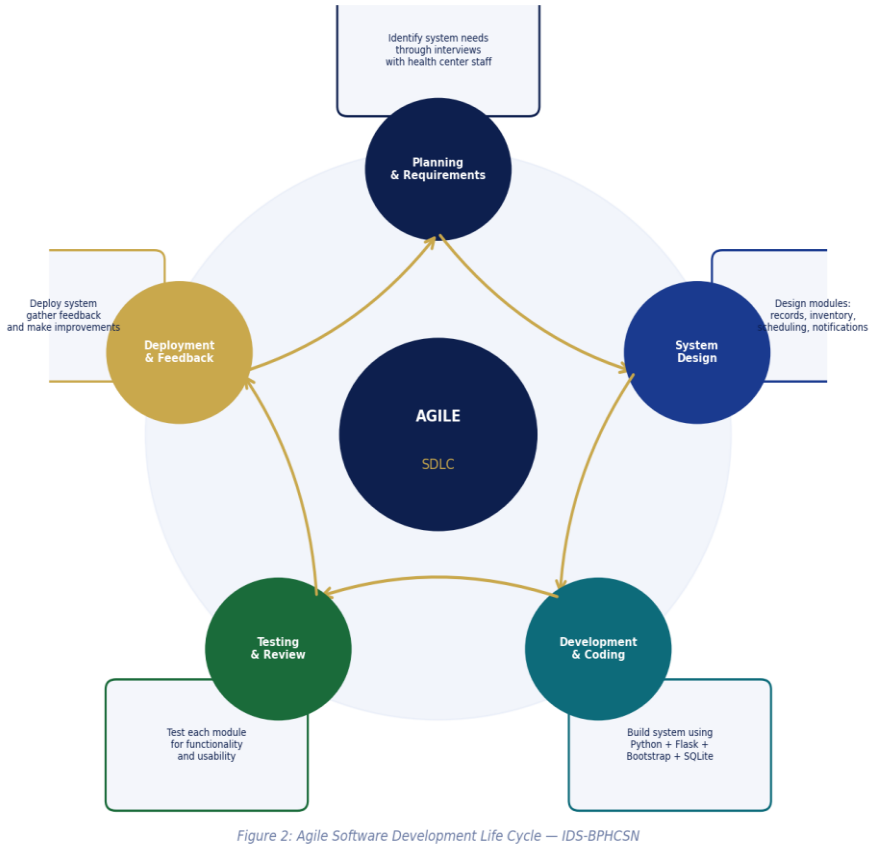


Figure 2: **Agile Software Development Life Cycle**

Phase 1: **Planning and Requirements**

The development team conducted initial consultations with health center staff and barangay officials of Barangay Poblacion to gather system requirements. These sessions focused on identifying the essential features of the system, including patient records management, medicine inventory monitoring, appointment scheduling, and community health notification via web and SMS. User roles were also defined Admin, Health Center Staff and Barangay Official to guide the system's access control and interface design.

Phase 2: **System Design**

Based on the gathered requirements, the developers designed the overall architecture and user interface of the IDS-BPHCSN. This phase covered the planning of the database schema, the layout of each module's interface, and the logical flow of data across the system. The Flask web framework was selected for the backend, SQLite for the database, and Bootstrap for the frontend interface, providing a lightweight and accessible technology stack appropriate for the capstone project scope.

Phase 3: **Development and Coding**

During this phase, the actual coding and implementation of the four core system modules took place patient records management, inventory monitoring, scheduling and notifications, and report generation. The development team

utilized AI-assisted code generation tools to accelerate development, as endorsed by the project's technical adviser. Role-based access control, form validation, and database operations were implemented progressively, with testing conducted after each module was completed.

Phase 4: **Testing and Review**

Each module was tested individually and as part of the integrated system to verify that all features performed their intended functions correctly. Testing covered functional testing verifying that data encoding, retrieval, alerts, and notifications worked as expected and usability testing with health center staff to ensure the interface was easy to navigate. Any bugs or usability issues discovered during this phase were documented and addressed before proceeding to the next development cycle.

Phase 5: **Deployment and Feedback**

The final phase involved deploying the IDS-BPHCSN on a local development server for demonstration and user acceptance evaluation. Feedback gathered from health center staff and barangay officials during this phase was used to make final refinements to the system's features and interface. The system was made accessible via a standard web browser on both desktop and mobile

devices, ensuring it could be used effectively on the hardware available at Barangay Poblacion Health Center.

System Requirements

The following are the minimum hardware and software requirements necessary to ensure the efficient performance of the proposed IDS-BPHCSN. These specifications are intended to support smooth system operations, data processing, and user interactions across both the administrator and client environments. Adhering to these requirements will help maintain system reliability, responsiveness, and compatibility throughout its implementation and usage.

Table 1

Software Specification

Admin Side

Software / Framework	Specification
Operating System	Windows 10 Windows 11 or higher
Backend Framework	Python 3.11 with Flask 3.0

Database	SQLite (built-in with Python)
Code Editor	Visual Studio Code (Version 1.88 or higher)
Version Control	Git/GitHub (Latest Version)
SMS Gateway	Semaphore API (Free Developer Tier)
Web Browser	Google Chrome/Microsoft Edge (Latest Version)

Client Side

Software / Framework	Specifications
Operating System	Windows 10 Windows 11 or Android (for mobile access)
Web Browser	Google Chrome/Microsoft Edge/Safari (Latest Version)

Bootstrap	Version 5.3 accessed via CDN no installation
JavaScript	Vanilla JS jQuery (CDN-based, no installation required)

Table 2

Hardware Specifications

Admin side

Component	Specification
Processor	Intel Core i5/AMD Ryzen 5 or higher
Memory (RAM)	8 GB—16 GB
Storage	512 GB SSD
Operating System	Windows 10 Windows 11 (64-bit)
Internet Connection	Minimum 10 Mbps stable broadband connection

Input Devices	Standard Keyboard and mouse

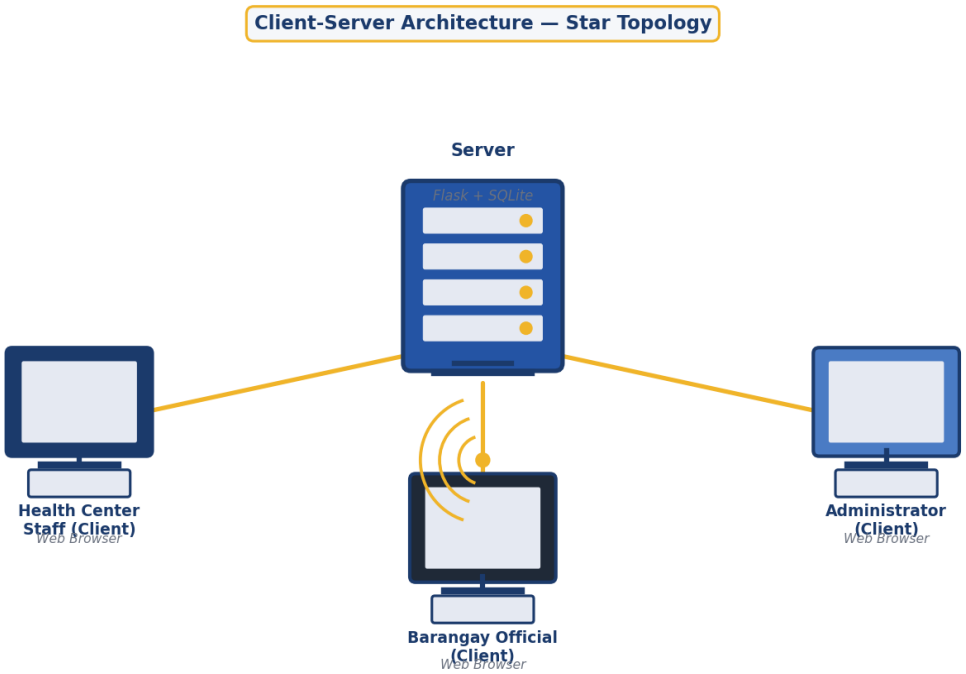
Client side

Component	**Specification**
Device	Desktop computer, laptop, or mobile device with a web browser
Processor	Intel Core / AMD Ryzen 3 or equivalent
Memory (RAM)	4GB-8GB
Storage	128 GB 256 GB SSD
Internet connection	Minimum 10 Mbps, optional mobile hotspot backup

Network equipment	Wi-Fi router with stable connection to the Flask web server

System Architecture

The IDS-BPHCSN follows a Client-Server Architecture using a Star Topology during both development and deployment phases. This structure centralizes all database operations and application logic on the Flask web server, while health center staff, barangay officials, and the administrator access the system through standard web browsers on their desktop computers or mobile devices, eliminating the need for any software installation at the client side.



Process model and data design

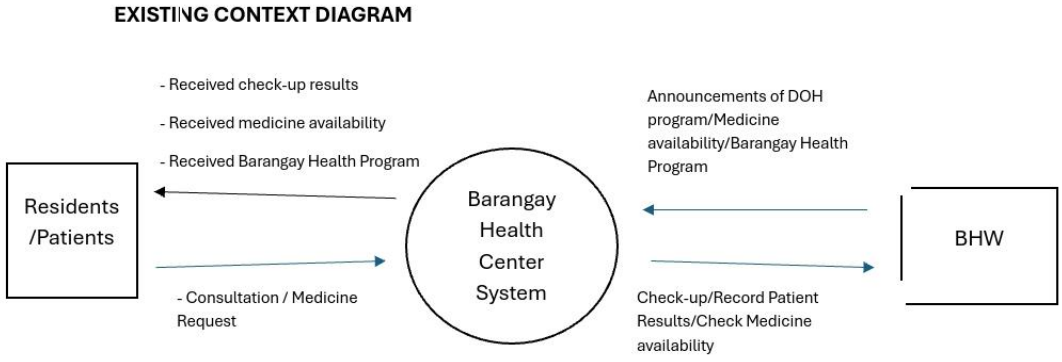


Figure 3 — Existing Context Diagram.

Figure 3 illustrates the context of the existing manual system currently used by Barangay Poblacion Health Center. Health center staff manually record patient consultations in paper logbooks, medicine inventory is tracked on handwritten sheets, and barangay officials disseminate health announcements through physical posting and word-of-mouth. This context diagram shows the absence of a centralized digital information flow between the health center and the community, highlighting the operational gaps that IDS-BPICSN is designed to address.

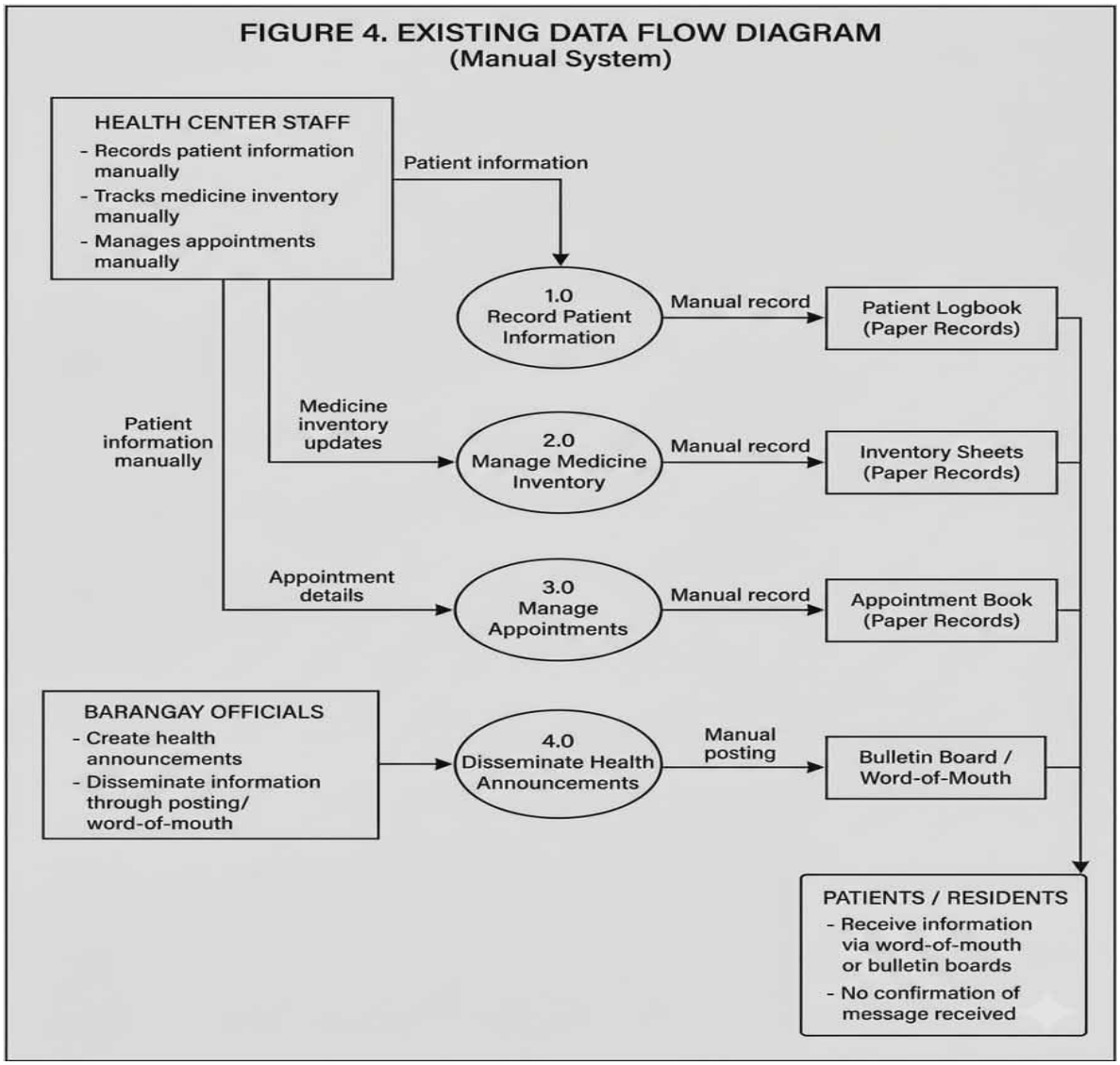


Figure 4 — Existing Data Flow Diagram

Figure 4 illustrates how data flows within the existing manual health center system. Patient information is manually recorded by staff into paper logbooks stored in physical folders, medicine usage is noted on inventory sheets without

automated alerts, and health announcements are posted physically on bulletin boards with no confirmation that residents have received the information. This data flow highlights the inefficiencies and communication gaps that the IDS-BPHCSN in designed to resolve.

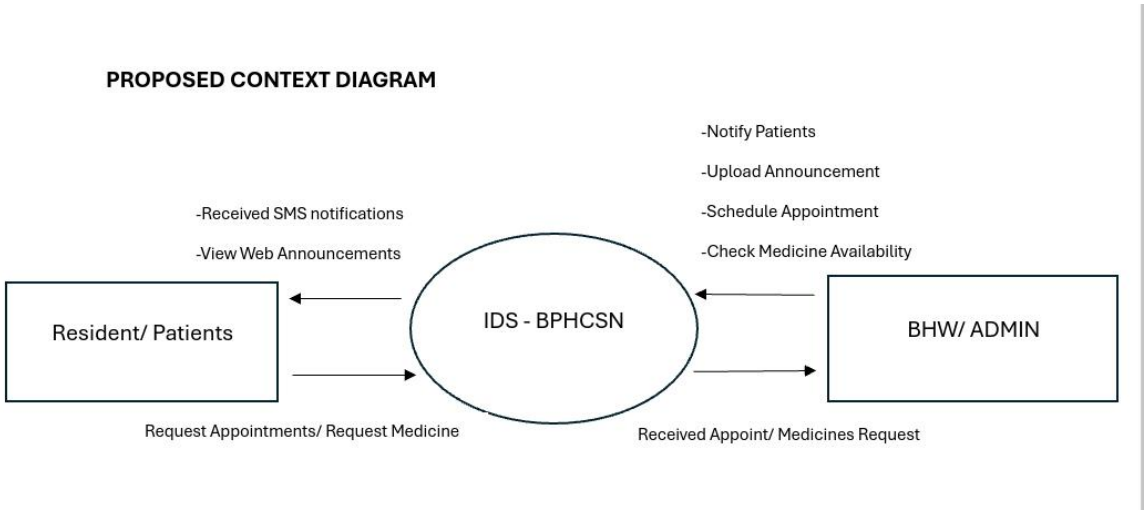


Figure 5 – Proposed Context Diagram

The Proposed Context Diagram illustrates how four external entities health center staff, barangay officials, patients, and community residents interact with the centralized IDS-BPHCSN web-based system. Health center staff submit patient records, medicine updates, and appointment schedules, while barangay officials post and broadcast health announcements. Residents receive health notifications through the system's SMS and web notification features.

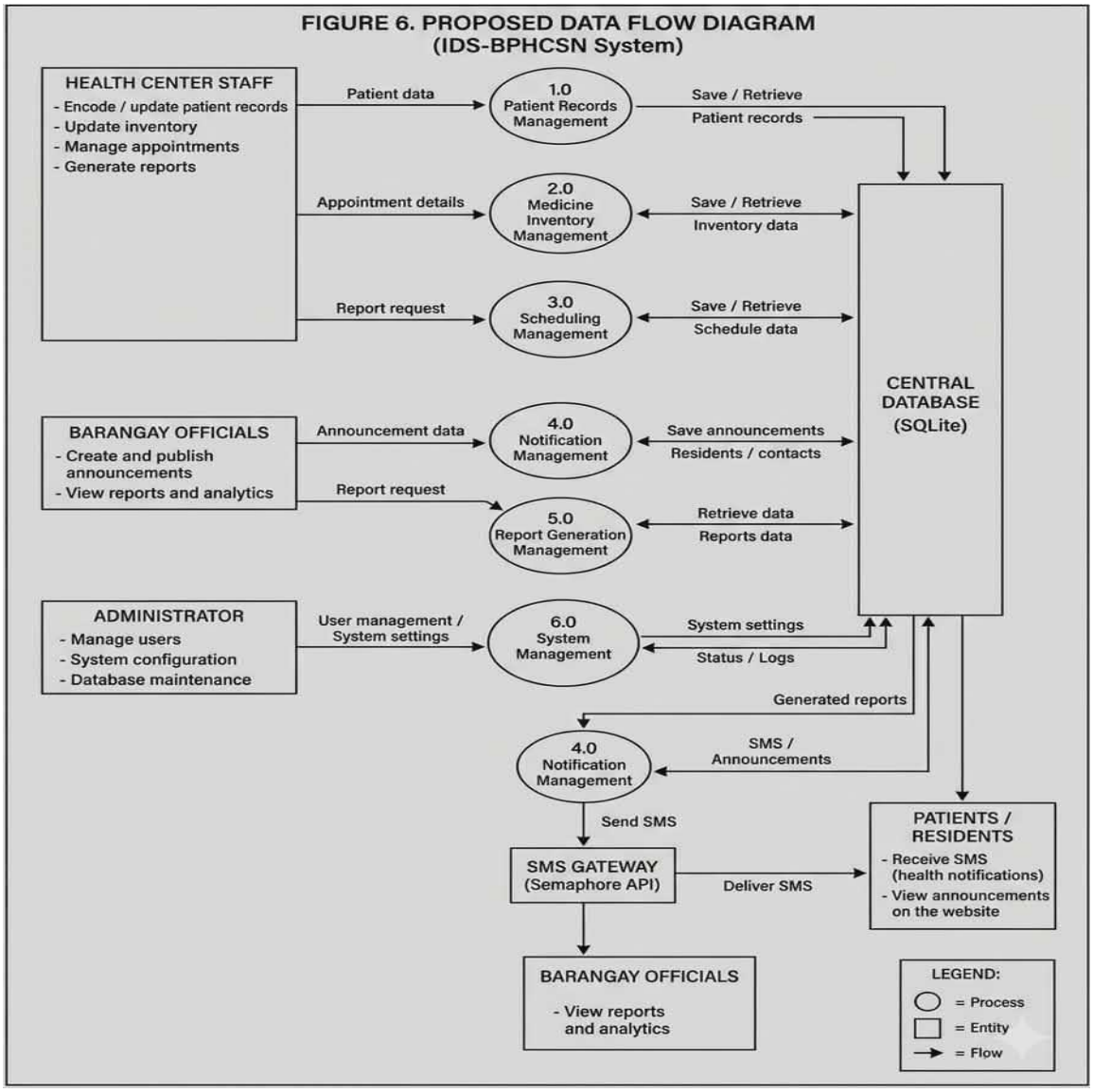


Figure 6 – Proposed Data Flow Diagram

The Proposed Data Flow Diagram illustrates the detailed data flow within the IDS-BPHCSN Authorized staff encode and retrieve patient records through the Patient Records module, while the Inventory module receives stock updates and generates low stock alerts. The Scheduling module manages appointments, the

Notification module dispatches announcements via Semaphore SMS, and the Report Generation module queries the database to compile health service summaries and inventory status reports.

Project Schedule

The project schedule presents the timeline and activities involved in the development of the IDS-BPHCSN, organized using a Gantt Chart. The chart outlines each development phase from requirements gathering and system design through module development, Testing, and final deployment along with their corresponding durations and target completion dates.

Project Schedule

Integrated Digital System for Barangay Poblacion Health Center Services and Community Notifications (IDS-BPHCSN)

Completed | In Progress | Not Started

Task No.	Task Name	Month – 2025											
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1.0	**Research & Documentation**												
1.1	Title Identification												
1.2	Title Approval												
1.3	Chapter 1–3 Writing & Revision												
2.0	**Phase 1: Planning & Requirements**												
2.1	Consultation with Stakeholders												
2.2	System Requirements Gathering												
2.3	User Role Definition												
3.0	**Phase 2: System Design**												
3.1	Database Schema Design												
3.2	UI/UX Design & Prototyping												
3.3	Technology Stack Finalization												
4.0	**Phase 3: Development & Coding**												
4.1	Patient Records Management Module												
4.2	Medicine Inventory Module												
4.3	Scheduling & Notification Module												
4.4	Report Generation Module												
5.0	**Phase 4: Testing & Review**												
5.1	Individual Module Testing												
5.2	Integration Testing												
5.3	Usability Testing with Staff												
6.0	**Phase 5: Deployment & Feedback**												
6.1	System Deployment on Local Server												
6.2	User Acceptance Evaluation												
6.3	Final Refinements & Submission												

Project Duration: January 2025 – December 2025 (2nd Week) | Methodology: Agile Software Development Life Cycle (SDLC) | Subject started: January 25, 2025 | Title approved: February 22, 2025 | Chapters 1–3 completed: April 7, 2025

Revenue Streams

The proposed IDS-BPHCSN for Barangay Poblacion Health Center may generate income through a combination of software licensing, optional customization, and value-added services such as technical support and training. These revenue streams are designed to ensure the long-term sustainability and maintenance of the system while offering flexible support options to the barangay. The table below outlines the key sources of revenue applicable for the system's deployment at the barangay level.

Revenue Source	Description
Software Licensing Fee	A one-time licensing fee for the IDS-BPHCSN paid by Barangay Poblacion upon final delivery and system turnover.
Customization Fee	Optional one-time charge for additional features or modules beyond the original system scope.
Support & Maintenance	Optional annual fee for continued technical support, system updates, bug fixes, and database backups (P3,000/year).
Training Service	Optional user training for health center staff and barangay officials on operating the IDS-BPHCSN (P1,000-P2,000/session).